Accidental Liberalization: Inflation, Specific Tariffs, and US Import Growth
PRELIMINARY AND INCOMPLETE

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Abstract

Unfortunately for trade economists, there is no pre-existing database of legislated US tariffs for the 1970s and 1980s prior to the current Harmonized System (HS) of product classification. We detail an algorithm to estimate the legislated ad valorem and specific tariffs applied by the US between 1972 and 1988 and show this algorithm performs remarkably well. Using simple trade theory, we also show how pass through of foreign price shocks to US import prices depends on the presence of specific tariffs. In turn, we combine our US tariff estimates with disaggregated foreign price shocks to quantify the effects on US trade patterns, and other US economic variables of interest, due to inflation-eroded specific tariffs and legislated cuts to ad valorem and specific tariffs. Inflationary foreign price shocks between 1972 and 1979 generated greater liberalization of US ad valorem equivalent tariffs than Tokyo Round in the 1980-1988 period. This "accidental" trade liberalization slashed US import tariffs with nearly every country, but the magnitude of the effect depended on the extent to which those countries exported goods facing by specific tariffs. Perhaps surprisingly, the Tokyo round of GATT initially restored US tariff protection by converting these specific tariffs to higher ad valorem levels before the Tokyo Round precipitated a scheduled phase out of both specific and ad valorem tariffs.

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1 Introduction

In analyses of the effects of trade policy, the most commonly used approach relates changes in trade flows or other market outcomes to changes in tariff rates. In the context of the United States, researchers have overwhelmingly applied such an approach to the period beginning in the late 1980s, despite the fact that tariff levels were already quite low by this period. There are likely two reasons for this for this focus. First, the vast majority of existing US preferential trade agreements were signed in or after 1989, beginning with the Canada-US Free Trade Agreement (CUSFTA).[1] In particular, the 1994 North American Free Trade Agreement (NAFTA) and China's accession to the World Trade Organization in 2001 have garnered substantial attention. Second, there is a lack of readily accessible product-level tariff data for years prior to 1989, when the US transitioned from the Tariff Schedule of the USA (TSUSA) to the Harmonized System (HS) as a tariff classification scheme. However, by confining focus to the last 35 years, scholars forgo the opportunity to explore other, larger liberalizations. Notably, between 1972 and 1989 the average tariff on US dutiable imports fell by one-third. As this is also a period of considerable growth in imports and labor market turmoil, the relative lack of work exploring the effects of tariff liberalization in this era represents a considerable missed opportunity.

In this paper, we attempt to help fill this gap in the literature. Using data on total duties collected at the country-product level, we develop an algorithm that estimates the ad valorem and specific tariffs specified by policy between 1972 and 1988. We also use simple trade theory to show how the pass through of foreign price shocks to US import prices depends on the extent of specific tariffs. In turn, we can show not only how liberalization affects import growth but also how specific versus ad valorem tariff liberalization differentially affect import growth. With our estimates of the US tariffs specified by policy from 1972-1988, we document a number of stylized facts.

First, the role of specific tariffs is considerable and varies dramatically over our sample period. The importance of specific tariffs within the US tariff schedule is captured by the share of product-level tariff revenue generated by specific tariffs – either in the aggregate or averaged across products – in the 1972-1979 period being very stable at around 30%. However, the Tokyo Round substantially dampens their importance with the share of tariff revenue generated by specific tariffs falling by about one-third upon implementation of the Tokyo Round in 1980. Interestingly, this picture is very similar when looking at the share

[1] The US-Israel Free Trade Agreement of 1985 is the lone exception.

of products with specific tariffs, which again falls by about one-third, and illustrates the considerable large-scale conversion of specific tariffs to ad valorem tariffs by the US in the Tokyo Round.

Second, the extent of liberalization in this era is substantially larger than would be suggested by an aggregate ad valorem equivalent. In particular, the aggregate ad valorem equivalent tariff for the US over the 1972-1988 period fell nearly 1.5% points from just under 5.5% to just over 4%. However, liberalization among a "balanced panel" set of country-products exporting to the US over our sample period reveals a much larger liberalization of around 2.5%-3% points which gets masked in the aggregate by changes in the composition imported products over time.

Third, US ad valorem equivalent tariffs – at an aggregate or disaggregate level – actually fell more in the 1972-1979 pre-Tokyo Round period than the 1980-1988 period that began with implementation of Tokyo Round tariff cuts. Moreover, with legislated specific and ad valorem tariffs essentially unchanged in the 1972-1979 pre-Tokyo Round period, all of the pre-Tokyo Round liberalization stems from inflationary foreign price shocks eroding the protection afforded by US specific tariffs. On average, annual product-level foreign price are around 10% in the pre-Tokyo 1972-1979 period but fall to around 2% for the post-Tokyo Round 1980-1988 period. At the more aggregate 2-digit SITC level, this leads to median cumulative inflation of around 75% during the 1972-1979 period. In turn, the ad valorem equivalence of US specific tariffs falls dramatically. The mean ad valorem equivalent specific tariff falls by over 50% from around 7% in 1972 to around 3% in 1979 and remains around 3% by the end of our sample in 1988.

Fourth, combining simple trade theory with our tariff estimates, we show how the decline in the use and size of specific tariffs notably changes the pass through elasticity of foreign price shocks to US import prices. With purely ad valorem tariffs, a 1% foreign price shock passes through fully as a 1% price shock to the US import price because the tariffs collected rise proportionately with an ad valorem tariff. However, since the specific tariff duties collected are invariant to the foreign price level, a 1% foreign price shock passes through at less than 1% to the US import price in the presence of specific tariffs. In 1972, the mean pass through elasticity of the foreign price shock to the US import price was around 94% but with many products having much lower pass through elasticities – various types of raw material products and clothing and apparel products had a pass through elasticity in the 20%-50% range. Across the board these pass through elasticities rise dramatically by 1988

with the much lower importance of specific tariffs in the US tariff code. The mean pass through elasticity is now 98% and the number of products with a pass through elasticity below 80% has fallen more than 80%.

The paper proceeds as follows. Section 2 describes the data, how we estimate US tariffs during the 1972-1988 period, and the accuracy of our tariff estimates. Section 3 derives the theoretical relationship between specific tariffs and the pass through elasticity of foreign price shocks to US import prices. This section also describes the stylized facts that emerge from our analysis. Section 4 concludes.

2 Data

Currently, no dataset contains product-level US tariffs for the period 1972-1988. Product-level ad valorem equivalent rates for this period have been calculated in previous work by dividing duties collected by import values. However, such an approach ignores the distinction between tariffs specified in ad valorem (percentage) terms and those specified in specific (per-unit) terms. As we document below, this distinction is empirically relevant during this era. To address this shortcoming, we construct a product-level dataset of US tariffs using a novel estimation procedure that captures both the ad valorem and specific components of tariffs specified by US tariff legislation.

An obvious alternative to our approach would be to digitize the annual US tariff schedules directly from available PDFs. However, this approach has two main drawbacks. First, it is time-intensive and subject to at least some degree of human error. Second, as we detail below, the unit of quantity is inconsistently recorded between annual US tariff schedules and US import data for a non-trivial number of observations. Our estimation procedure allows us to agnostically identify these inconsistencies – something that could not be done if one were to digitize the tariff schedule manually. A key consequence of failing to recognize these inconsistencies is incorrectly calculating the ad valorem equivalent tariff for products with specific tariffs by orders of magnitude.

2.1 Estimating tariffs 1972-1988

The key data we use to estimate tariffs is published by the US Census Bureau in the US Annual Import Data Bank (IDB). We obtain the IDB data from The Center for International Trade Data.[2] These data include total duties collected as well as import values, dutiable values, and quantities. The data are highly disaggregate: each observation represents a TSUSA 7-digit product j exported by country i to a US port d in year t, subject to tariffs defined by rate provision code r.[3,4]

Table 1 describes the two-digit rate provision codes in the data. The first digit describes

[2] Feenstra (1996) aggregates these data to the country-product-year level.

[3] Since 1989, the US has used the globally recognized Harmonized System (HS) to define internationally traded products. Prior to 1989, the US used its own product classification system known as the Tariff Schedule of the USA (TSUSA). The data record trade flows at the 7-digit TSUSA level, though the US tariff schedule defined tariffs at the 5-digit level.

[4] To streamline notation, we refer to the port of entry-by-port of unlading pair as port d.

whether the tariff is duty free, specific, ad valorem, or compound (i.e., has both specific and ad valorem components). The second digit describes whether the tariff is the MFN normal trade relations (NTR) tariff from column 1 of the US tariff schedule or the non-NTR tariff from column 2. The vast majority of imports fall under one of these eight potential rate provision codes. For example, rate provision code 22 indicates a non-NTR specific tariff, while rate provision code 31 indicates an NTR ad valorem tariff. The codes 5*, 7*, *3, *4 and *8 capture various infrequently used types of tariffs.

First Digit	Duty Type	Second Digit	Rate
1*	Duty Free	*1	Col 1
2*	Specific Duty	*2	Col 2
3*	Ad Valorem	*3	Col. 1 Exceptions
4*	Compound	*4	Col. 2 Exceptions
5*	Minimum	* 8	Cuba Special
7*	Special		

Table 1: Rate provision codes (r)

We estimate 7-digit TSUSA tariffs annually using a two-step iterative procedure that revolves around the definition of duties collected. Equation (1) defines total duties, TD, collected from country i exporting 7-digit TSUSA product j to US port d in year t and facing rate provision code r:

$$TD_{idjrt} = \tau_{jrt} DV_{idjrt} + f_{jrt} Q_{idjrt} + \epsilon_{idjrt}. \tag{1}$$

Here, τ_{jrt} and f_{jrt} are, respectively, the ad valorem and specific tariff components, while DV and Q represent dutiable import value and the physical quantity of imports. ϵ_{idjrt} represents measurement error in the data.

Absent measurement error, one could exactly *calculate* the unobservable tariffs τ_{jrt} and f_{jrt} given observed data on duties, import values, and quantities. Consider a particular product j facing rate provision code r in year t. One could then calculate:

$$\tau_{idjrt} = \frac{TD_{idjrt}}{DV_{idjrt}} \text{ for } r = 2^* \tag{2}$$

$$f_{idjrt} = \frac{TD_{idjrt}}{Q_{idjrt}} \text{ for } r = 3^* \tag{3}$$

$$TD_{idjrt} = \tau_{idjrt} DV_{idjrt} + f_{idjrt} Q_{idjrt} \text{ for } r = 4^*. \tag{4}$$

Naturally, one would need two exporter-port observations to calculate τ_{idjrt} and f_{idjrt} for compound tariffs since (4) is a single equation with two unknowns τ_{idjrt} and f_{idjrt}.[5,6] Of course, in the absence of measurement error, the calculated values of τ_{idjrt} and f_{idjrt} would be identical for all exporter-port pairs (i.e. i-d pairs). That is, equations (2)-(4) actually yield the MFN tariffs τ_{jrt} and f_{jrt} in (1) for any exporter-port observations used to solve equations (2)-(4).

Two features of the IDB data complicate our analysis. First, the data can report two separate (and different) quantity values that correspond to different units of quantity measurement (e.g. pounds, gallons, inches, count). Nevertheless, second, the data do not record the unit of quantity measurement. When the data report two separate quantity values, one would need two exporter-port observations to calculate the correct f_{idjrt} for products that only have specific tariffs ($r = 3$). In this case, one would calculate a value of f_{idjrt} using (3) for each unit of the two units of quantity. The correct f_{idjrt} would be the the one that is constant across the two exporter-port observations.[7] Similarly, one would need two sets of exporter-port observation pairs to calculate the correct τ_{idjrt} and f_{idjrt} for products with compound tariffs ($r = 4$). In this case, one would separately solve the simultaneous equations given by (4) for each unit of quantity. Only one unit of quantity would give a constant τ_{idjrt} and f_{idjrt} across the two sets of exporter-port observation pairs and this would be the correct τ_{idjrt} and f_{idjrt}.

Further, measurement error is pervasive in the IDB data. It can arise for various reasons, including data censoring and rounding, data entry errors, and changes in tariff rates within a given year.[8] Such measurement error implies that we can only estimate the MFN tariffs τ_{jrt} and f_{jrt} with noise using the procedure just described. Nevertheless, the above procedure forms the basis of our estimation procedure.

We use a two-step iterative estimation procedure: the first step provides initial estimates of the tariffs, while the second refines estimates by removing estimates exhibiting large errors.

[5]Additionally, it would require that the two versions of equation (4) defined by the two exporter-port pairs are linearly independent or, in other words, define two non-parallel lines in τ-f space.

[6]Since each exporter faces the same tariff for a given product j, rate provision r, and year t, these two exporter-port observations could either be for (i) the same exporter and different ports or (ii) the same port and different exporters.

[7]This requires that the percentage change in the quantity amount across the two exporter-port observations in the data is different for each of the two units of quantity. This is satisfied in general as long as the two units of quantity in the data are not scalars of each other (e.g. pounds and ounces).

[8]For example, the calculated duties from China's exports to the US in 1980 reflect a combination of the Column 2 rates in January and February prior to receiving NTR status but the Column 1 rates for the rest of year after obtaining NTR status.

For products that have only ad valorem tariffs ($r = 3^*$) or specific tariffs ($r = 4^*$), we estimate τ_{jrt} and f_{jrt} as a weighted average of τ_{idjrt} and f_{idjrt}, as defined by (2)-(3), across exporters i and ports d:

$$\hat{\tau}_{jrt} = \sum_{i,d} \tau_{idjrt} \cdot \omega^{DV}_{idjrt} \tag{5}$$

$$\hat{f}^m_{jrt} = \sum_{i,d} f^m_{idjrt} \cdot \omega^{Q,m}_{idjrt}. \tag{6}$$

Given quantity amounts for specific tariff products reported under two separate units, we estimate $\hat{f}^m_{jrt}$ separately for each unit of quantity measurement m. We then define our estimate $\hat{f}_{jrt}$ as the $\hat{f}^m_{jrt}$ for the unit of measurement m with the smallest weighted mean squared error $\sum_{i,d} \omega^{f,m}_{idjrt} \left(\hat{f}^m_{ipjrt} - \hat{f}^m_{jrt}\right)^2$. The specific tariff estimate weights are the quantity weights $\omega^{Q,m}_{idjrt} = \frac{Q^m_{idjrt}}{\sum_{i,d} Q^m_{idjrt}}$ and the ad valorem tariff estimate weights are the dutiable value weights $\omega^{DV}_{idjrt} = \frac{DV_{idjrt}}{\sum_{i,d} DV_{idjrt}}$.

Intuitively, we follow a similar approach when estimating compound tariffs. For a given unit of measurement m, we obtain estimates $\hat{\tau}^m_{jrt}$ and $\hat{f}^m_{jrt}$ by estimating equation (1) using ordinary least squares where the sample is the set of exporters i and ports d associated with the a given cluster of product-by-rate-provision-by-year (i.e. v-by-r-by-t) observations. When the data records two units of measurement, we again define $\hat{\tau}_{jrt}$and $\hat{f}_{jrt}$ as the $\hat{\tau}^m_{jrt}$ and $\hat{f}^m_{jrt}$ for the unit of measurement m with the smallest weighted mean squared error $\sum_{i,d} \omega^{DV}_{idjrt} \left(TD_{idjrt} - \hat{TD}^m_{idjrt}\right)^2$.

After obtaining initial tariff estimates $\hat{\tau}_{jrt}$ and $\hat{f}_{jrt}$ for all 7-digit TSUSA products, we move to the second step of our estimation procedure. Measurement error for an individual observation can be very large, such as when countries shift from Column 2 to Column 1 tariffs during the year. By iteratively removing observations with substantial measurement error, we improve the accuracy of our tariff estimates.

To this end, we begin by defining ad valorem equivalent tariffs in the data, τ^{AVE}_{idjrt}, and our estimate of these, $\hat{\tau}^{AVE}_{idjrt}$:

$$\tau^{AVE}_{idjrt} = \frac{TD_{idjrt}}{DV_{idjrt}} \text{ and } \hat{\tau}^{AVE}_{idjrt} = \hat{\tau}_{idjrt} + \frac{\hat{f}_{jrt} Q_{idjrt}}{DV_{idjrt}}. \tag{7}$$

For relatively small ad valorem equivalent tariffs, the log estimation error

$$e_{idjrt}^{Est} = \ln\left[p_{idjrt}^{*}\left(1+\tau_{idjrt}^{AVE}\right)\right] - \ln\left[p_{idjrt}^{*}\left(1+\hat{\tau}_{idjrt}^{AVE}\right)\right] \tag{8}$$

approximates the percentage difference between the actual value and our estimated value of the tariff-inclusive price. The first iteration looks within every cluster of product-by-rate-provision-by-year observations and finds the exporter-port with the largest estimation error e_{ipjrt} and drops all such "largest error" observations as long as the error e_{idjrt} exceeds 0.001. By definition, this threshold tolerance is approximately 0.1 of the free-on-board price. Using the smaller sample, the second iteration re-estimates the tariffs following our "initial values" step above and finishes by dropping any further "largest error" observations. We stop this step of the iteration process after six iterations. As a subsequent step in the iteration, we drop the set of exporters in a given year where the mean error $\bar{e}_{it}$ across all product-by-port-by-rate-provision observations within a cluster of export-year observations exceeds 0.001.

2.2 Assessing accuracy of estimated tariffs

In general, we expect that estimation error will drive a wedge between estimated tariffs and the actual tariffs in the PDF files of the TSUSA. However, we find that this estimation error is extremely small in our dataset. Moreover, at least for products that have specific tariffs, our estimation error is substantially smaller than the error that would result from a *perfect* digitization of the tariff data from PDF files of the TSUSA.

Our first method for assessing the accuracy of our estimated tariffs is to compare the product-level tariff-inclusive price in the import data with our estimate of it. The log difference between these, which approximates the percentage difference for small values of the ad valorem equivalent tariff, was defined as the log estimation error e_{idjrt} in (8). Figure 1 shows the distribution of e_{idjrt}. The dashed red lines indicate thresholds of $|e_{idjrt}| = 0.001$ which, for relatively small values of ad valorem equivalent tariffs, represents an estimation error of approximately 0.1% of the tariff-inclusive price. Less than 0.01% of import value and only 2.8% of observations lie outside of these thresholds. Thus, Figure 1 clearly illustrates our tariff estimates are extremely accurate.

Our second method for assessing the accuracy of our estimation strategy is to compare our estimated tariffs with a small sample of digitized tariffs. To this end, we digitized Schedule 1 from the PDF file of the 1978 TSUSA. This represents about a 10% sample of

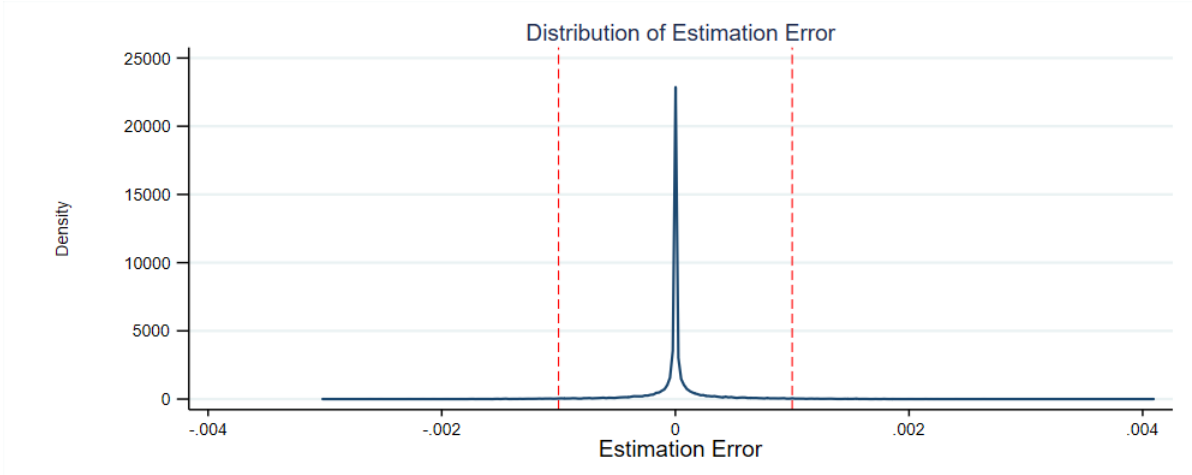

Figure 1: Distribution of estimation error

7-digit TSUSA tariff lines. Schedule 1 also over-represents agricultural products and, in turn, over-represents specific tariffs. Hence, it is a good sample to assess the accuracy of our estimation procedure because we expect greater measurement error for products with specific tariffs.

Table 2 presents the first x products with compound tariffs in Schedule 1 and shows both our estimated tariffs and the digitized tariffs. Compound tariffs are the most difficult tariffs to estimate given the fact that we need to estimate them via regression and their specific tariff component is complicated by the fact that quantity amounts can be recorded in two units of measurement in the data. Nevertheless, Table 1 shows the striking accuracy of our estimated tariffs. Only 1 product has anything approaching non-trivial estimation error: product 17624 has an estimation error of 0.002% points for the ad valorem tariff and 0.16 cents per unit for the specific tariff (on a digitized specific tariff of 1.13 cents per unit). In contrast, our estimated tariffs exactly match the digitized tariffs for the majority of products.

While our estimated tariffs may be extremely accurate, one may nevertheless reasonably assume that digitizing tariffs would be more accurate than estimating the tariffs. To this end, we define the log digitization error as

$$e^{Dig}_{idjrt} = \ln\left[p^*_{idjrt}\left(1+\tau^{AVE}_{idjrt}\right)\right] - \ln\left[p^*_{idjrt}\left(1+\hat{\tau}^{AVE}_{idjrt}\right)\right] \tag{9}$$

where $\hat{\tau}^{AVE}_{idjrt}$ was defined in equation (7) but now $\hat{\tau}_{idjrt}$ and $\hat{f}_{idjrt}$ are digitized rather than estimated. Again, for small values of ad valorem equivalent tariffs, the log digitization error approximates the percentage difference between the actual value and our digitized value of

Table 2: Sample of Estimated vs Digitized Tariffs

TSUSA	$\hat{f}_{jt}$	f_{jt}	$\hat{\tau}_{jt}$	τ_{jt}
14410	0.0498	0.0500	0.2504	0.2500
14412	0.0320	0.0320	0.1000	0.1000
14420	0.0320	0.0320	0.1000	0.1000
14699	0.0700	0.0700	0.1000	0.1000
15405	0.0698	0.0700	0.1004	0.1000
17065	1.0601	1.0600	0.0500	0.0500
17066	0.9500	0.9500	0.0500	0.0500
17072	1.9101	1.9100	0.1050	0.1050
17624	0.0097	0.0113	0.0520	0.0500
17655	0.0090	0.0090	0.0400	0.0400

the tariff-inclusive price.

Figure 2 plots the log estimation error e^{Est}_{idjrt} against the log digitization error e^{Dig}_{idjrt} for our 10% sample of digitized tariffs. A striking feature of this figure is the different scale of the axes. The dashed red lines on the x-axis how a log estimation error of $\left|e^{Est}_{idjrt}\right| = .001$ while the scale on the y-axis ranges up to a log digitization error of $e^{Dig}_{idjrt} = -4$ which is orders of magnitude larger. On one hand, Figure 2 shows that both estimation and digitization are extremely accurate. The large clump at the origin indicates that the vast majority of observations have essentially zero estimation error and zero digitization error. However, nearly all observations away from the origin lie in the various masses that essentially have an estimation error of zero but a log digitization error between -0.5 and -3. These are extremely large log digitization errors. Ultimately, Figure 2 illustrates that, at least for this 10% sample of 1978 TSUSA tariff lines, the estimated tariffs are notably more accurate than the digitized tariffs.

To understand the discrepancy between the estimated and digitized tariffs, Figure 3 breaks down the underlying source of discrepancy. As already shown in Figure 2, the vast majority of observations – around 94% – have essentially zero estimation and digitization error. While non-trivial estimation error – defined as $\left|e^{Est}_{idjrt}\right| > 0.001$ – happens rarely, it happens for 0.86% of observations. While manual digitization error also happens because we mis-transcribed the PDF, it is even more rare and accounts for 0.04% of observations. The surprising source of discrepancy between the estimated and digitized tariffs is the other 5% of observations where we have zero estimation error but non-trivial digitization error. These observations largely correspond to the mass of observations described in Figure 2 with zero

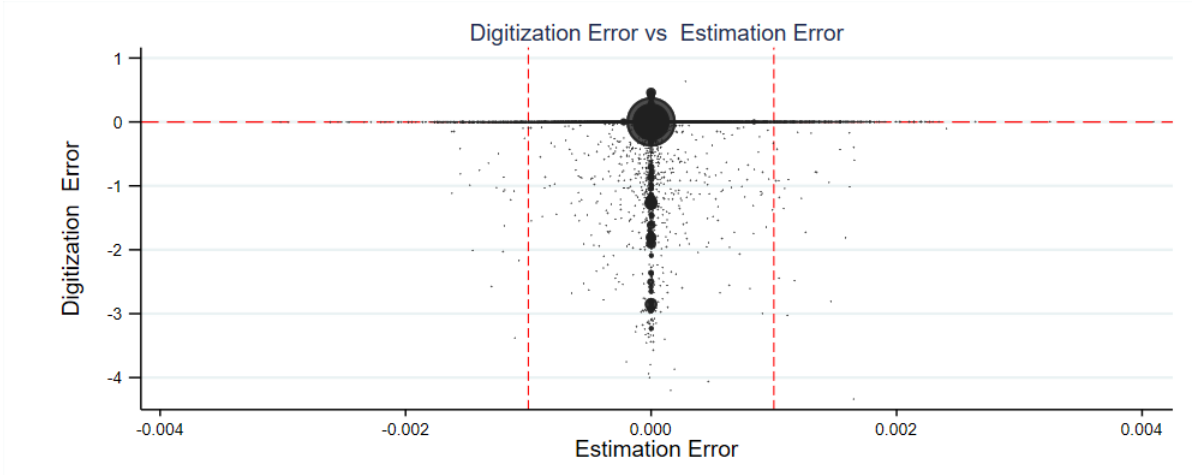

Figure 2: Scatterplot of estimation error versus digitization error

estimation error but log digitization errors between -0.5 and 3.

Fundamentally, the digitization error is not an error in terms of mistranscribing the PDF of the 1978 TSUSA or this PDF listing incorrect tariffs. Rather, it is a digitization error – a discrepancy between the digitized data from the PDF file of the TSUSA and the tariff duties data in Census IDB – that reflects the different unit of quantity measurements across the two data sources for a given product.

As an example, the PDF file of the 1978 TSUSA lists product 420.94.00 as "Sodium Chloride (in bulk)" with a tariff of \$0.008 cents per hundred weight (CWT, where a hundred weight equals 100 pounds). The 1978 IDB gives a unit value for this product of $p^* = \$0.55$. But, crucially, the IDB does not provide the quantity unit of measurement. Naturally, one would assume the unit is CWT to match the TSUSA. Under this assumption, the ad valorem equivalent of the specific tariff would be $\frac{\$0.008}{\$0.55} = 1.45\%$.

However, our estimation procedure reveals a specific tariff of $\frac{TD}{Q} = \$0.16$ per unit. Not only is this precisely 20 times the digitized tariff, but one tonne equals 20 CWT. Thus, the unit of measurement in the IDB appears to be tonnes rather than CWT. This may not seem problematic: since one tonne equals 20 CWT then a specific tariff of \$0.16 per tonne is equivalent to a specific tariff of \$0.008 per CWT. However, calculating the ad valorem equivalent of the specific tariff is problematic because the IDB does not record whether the unit of quantity measurement is tonnes or CWT. As discussed above, it seems natural to assume the IDB unit value is in terms of CWT when digitizing the tariffs which gives a digitized ad valorem equivalent tariff of $\frac{\$0.008}{\$0.55} = 1.45\%$. However, estimating the tariffs

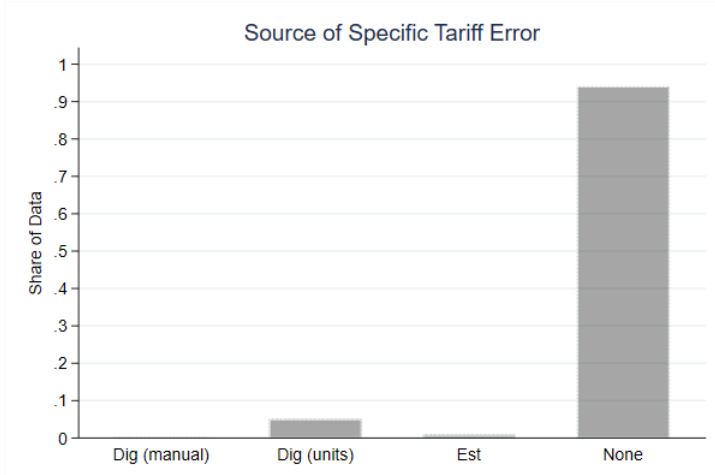

Figure 3: Sources of estimation and digitization error

reveals that the unit of quantity in the IDB is tonnes and hence the ad valorem equivalent is 20 times larger at $\frac{\$0.16}{\$0.55} = 29\%$. This discrepancy between the implicitly assumed unit of quantity measurement when digitizing the TSUSA tariffs and the revealed unit of quantity when estimating the tariffs using the IDB means that a *perfect* digitization of the TSUSA is notably less accurate than estimating the tariffs.

3 Stylized facts: US tariffs 1972-1988

3.1 Pass through of foreign price shocks

The US consumer price p_{ij} is related to the foreign price p^*_{ij} as

$$p_{ij} = p^*_{ij}\left(1 + \tau_{ij} + \frac{f_{ij}}{p^*_{ij}}\right) \equiv p^*_{ij}\left(1 + AVE_{ij}\right) \tag{10}$$

where τ_{ij}, f_{ij}, and $AVE_{ij} \equiv \tau_{ij} + \frac{f_{ij}}{p^*_{ij}}$ are, respectively, the ad valorem, specific and ad valorem equivalent tariffs imposed by country j on imports of variety i. Naturally, $\tau_{ii} = f_{ii} = 0$. The

pass through elasticity of a foreign price shock to the US consumer price is then

$$\begin{aligned}\eta_{ij} &= \frac{\partial \ln p_{ij}}{\partial p^*_{ij}} p^*_{ij} \\ &= \frac{\partial p_{ij}}{\partial p^*_{ij}} \frac{p^*_{ij}}{p_{ij}} \\ &= 1 - \frac{f_j}{p^*_{ij}(1+\tau_j) + f_j}.\end{aligned}$$

Intuitively, a 1% foreign price shock passes through as a 1% price shock to the US consumer price in the absence of specific tariffs. But, the presence of specific tariffs dampens the pass through of the foreign price shock and more so when the specific tariff accounts for a larger share of the US consumer price. Importantly, this pass through elasticity varies not only across products j but also across exporters i for a given product j because different exporters i can have different foreign prices p^*_{ij}. Given the pass through of the foreign price shock, the effect on US imports would depend on the trade elasticity and, hence, a particular model of trade that generates the model-specific trade elasticity.

To better understand the way that specific tariffs dampen the pass through elasticity, we can decompose the dampening effect into two separate terms. Note that

$$\begin{aligned}\frac{f_j}{p^*_{ij}(1+\tau_j) + f_j} &= \frac{f_j}{p^*_{ij}\tau_j + f_j} \frac{\tau_j + \frac{f_j}{p^*_{ij}}}{1 + \tau_j + \frac{f_j}{p^*_{ij}}} \\ &\equiv STS_{ij} \frac{AVE_{ij}}{1 + AVE_{ij}}. \end{aligned} \tag{11}$$

The first term on the right hand side, STS_{ij}, represents the specific tariff share of total tariff duties on product j collected by the US from exporter i. This will vary across products because different products have different specific tariffs f_j. Naturally, STS_{ij} only lies strictly between 0 and 1 for compound tariffs (i.e. products with both ad valorem and specific tariffs) because $STS_{ij} = 0$ for products with only ad valorem tariffs and $STS_{ij} = 1$ for products with only specific tariffs. The second term on the right hand side $\frac{AVE_{ij}}{1+AVE_{ij}}$ reflects the importance of overall protection in the price paid by US consumers. This varies across products because different products receive different levels of protection. Ultimately, specific tariffs dampen foreign price shock pass through when they account for a larger share of duties (i.e. higher STS) and protection accounts for a larger share of the US consumer price (i.e. higher $\frac{AVE}{1+AVE}$).

3.2 Price shocks

US headline inflation, and inflation across many parts of the world, was higher in the 1970s than higher than the rest of the 1900s, the 2000s and the 2010s. In general, part of this US inflation stems from rising prices of imported goods. The pass through elasticity of foreign price shocks quantifies the magnitude of this effect.

As a way to illustrate the foreign price shocks we observe in the data, we aggregate our exporter-product-port-year data across ports to the exporter-product-year level. An exporter-product annual price shock is then $\Delta \ln\left(p^*_{ij,t+1}\right) \equiv \ln\left(p^*_{ij,t+1}/p^*_{ijt}\right) = \ln\left(p^*_{ij,t+1}\right) - \ln\left(p^*_{ij,t}\right)$. This timing means price shocks in our graphs for 1972 are illustrating unit values in 1973 vis-a-vis 1972.

The left hand side of Figure 4 looks at the time series of foreign price shocks in our data. It does so by looking at the median, mean and import-weighted mean. The takeaway is that inflation on imported goods was much higher in the 1970s than the 1980s. This is emphasized by the right hand side of Figure 4 which pools the observations in the pre- and post-Tokyo periods of 1972-1979 and 1979-1980.[9]

Figure 5 shows how inflation varies across 2-digit SITC sectors and how these sector-specific differences vary between the pre- and post-Tokyo periods. We again see the stark difference in the magnitude and sign of annual price shocks in the 1970s vs 1980s. In the pre-Tokyo period, the the largest average annual price shocks are in sectors with very few imports (SITC 32 (Coal, coke & briquettes), SITC 60 (no description), SITC 70 (no description) and SITC 97 (Gold)). But, large sectors do see significant average annual inflation in the pre-Tokyo period. All of the following sectors see median annual inflation of around 10%: SITC 78 (road vehicles, 13% of US imports), SITC 84 (apparel, 4% of US imports), SITC 67 (iron & steel, 4% of US imports), SITC 68 (non-ferrous metals, 3% of US imports), and SITC 64 (paper, 2% of US imports).

Figure 6 shows the product-level mean or median log price change over the 7 year period 1972 to 1979. Thus, it reflects cumulative inflation in the pre-Tokyo period. The two largest US imports apparel (SITC 84) and petrol (SITC 33) see see the largest cumulative inflation of about 2.25 log points and 1.5 log points respectively (within-sector median). Other large

[9] Petrol (SITC 33) is the largest 2-digit SITC US import at over 20% of US imports and sees about 20% deflation in 1985 which helps explain the unusually large difference between the mean and median annual price shock in versus the import-weighted mean in 1985. A figure below will look the sectors driving the very high inflation for 1973 (which is actually inflation $t + 1 = 1974$ vs $t = 1973$ where the OPEC oil crisis begins in October 1973).

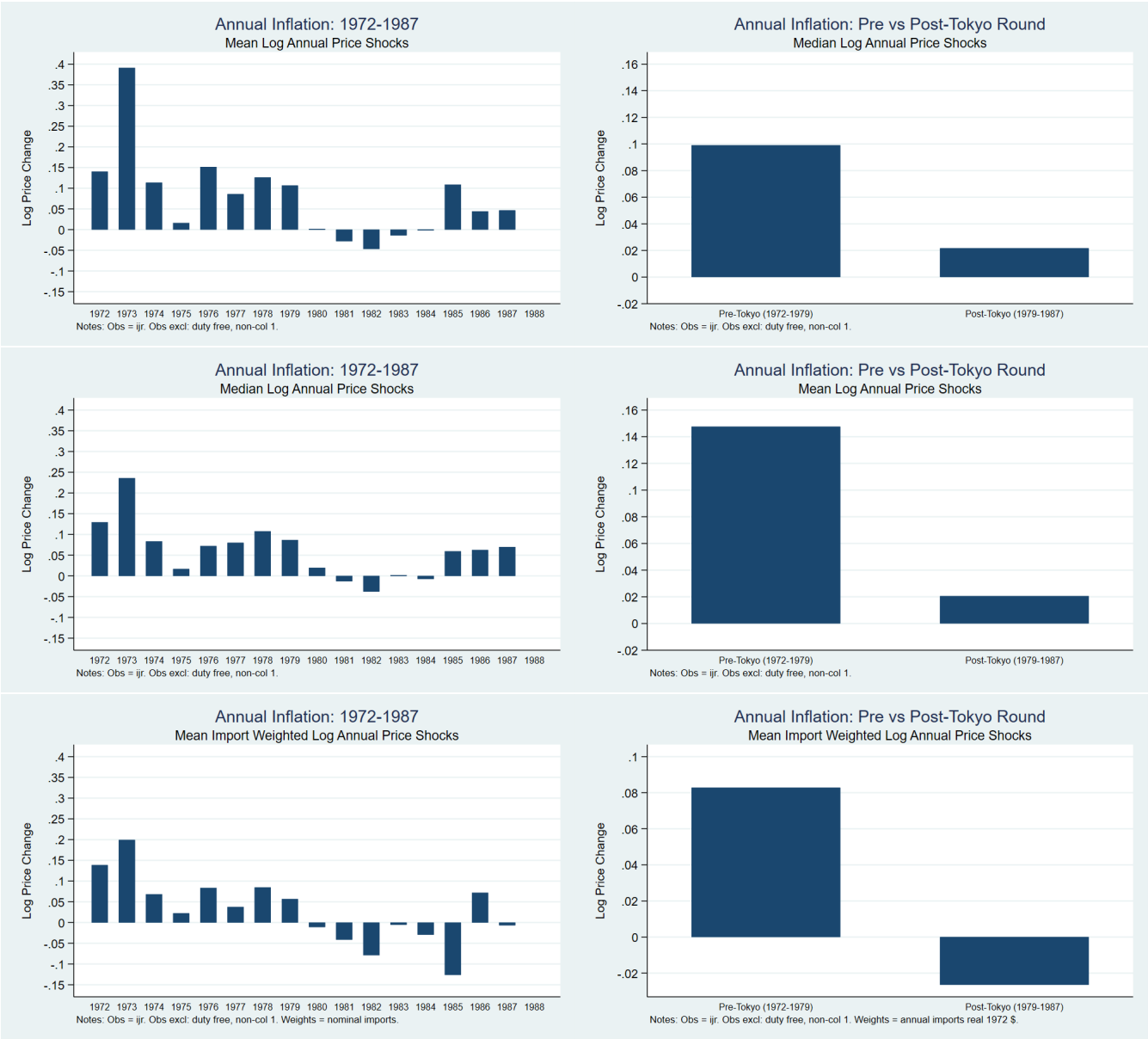

Figure 4: Annual product-level inflation

US imports with large cumulative inflation include metalliferrous ores (SITC 28, 1% of US imports) at around 2.25 log points and coffee, tea, cocoa and spices (SITC 7, 2% of US imports) at around 1.5 log points.

Naturally, such large inflation substantially erodes the ad valorem equivalence of specific tariffs. Regardless of whether we look at the median, mean or import-weighted mean, Figure 7 the AVE of the the specific tariff falls by at least 50% from 1972 to 1979 and remains

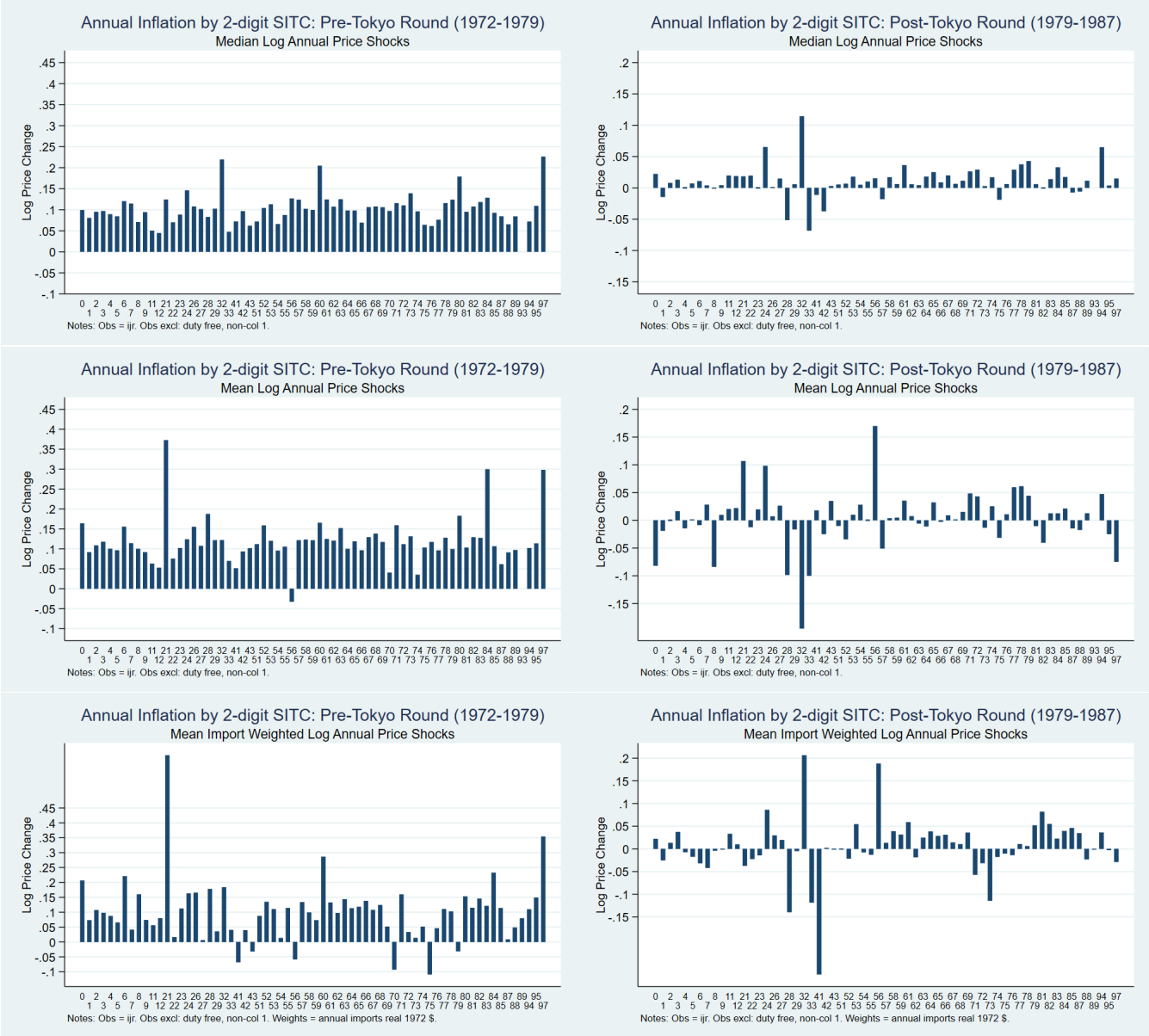

Figure 5: Annual product-level inflation by 2-digit SITC

relatively stable thereafter. The decline in the AVE of the specific tariff is cleanest when looking the median but more volatile when looking at the mean or the import-weighted mean. Figure 8 shows some examples of how product-specific inflation evolves over time and how product-specific inflation varies across sectors at a given point in time. Above we saw that 1973 inflation was unusually high. The first OPEC oil crisis beginning in October 1973 (and 1973 inflation in our data is $t+1 = 1974$ vs $t = 1973$) helps explain this. Indeed, petrol is the easily the largest US import at the 2-digit SITC level at 23% of US imports. And,

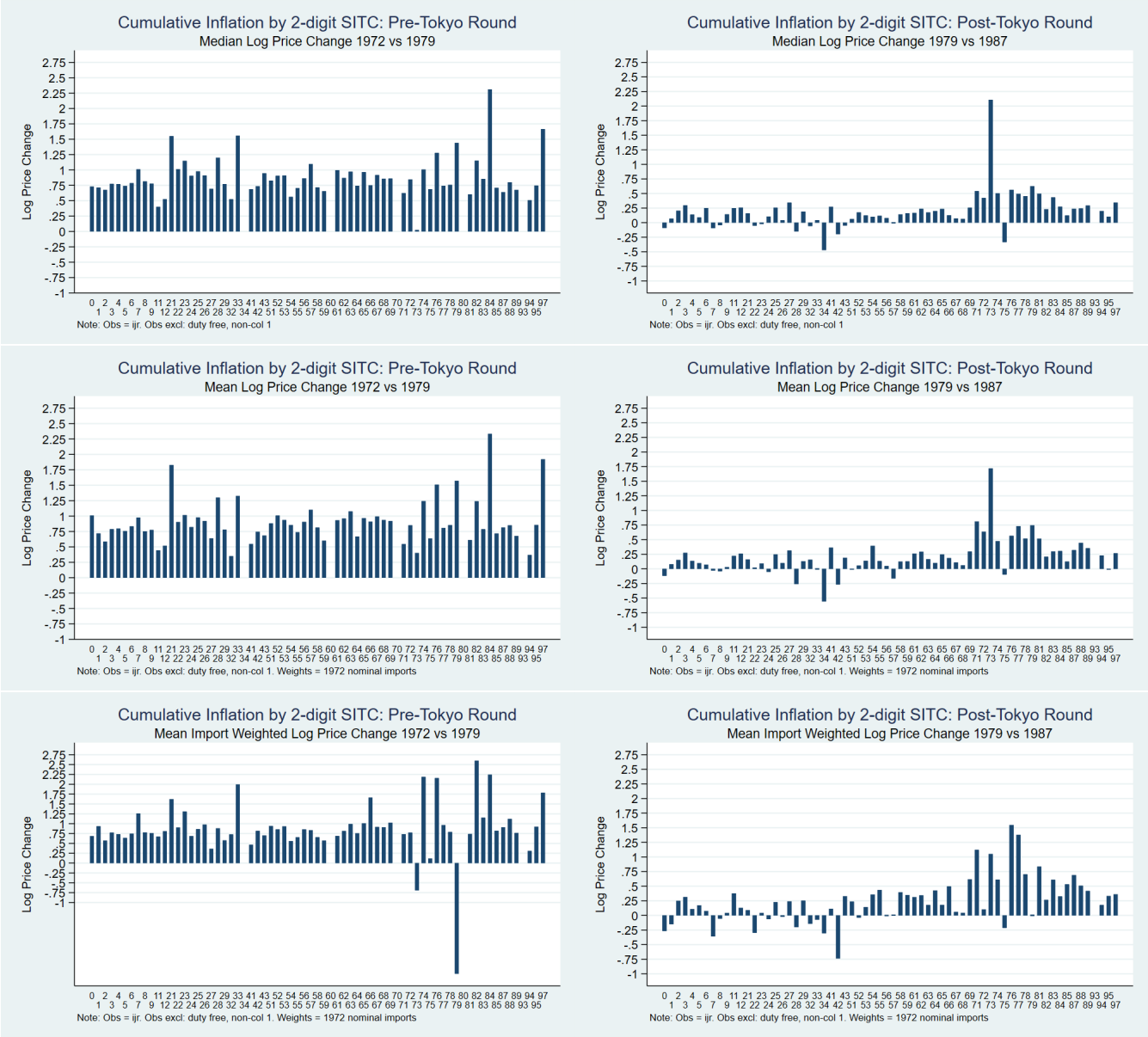

Figure 6: Cumulative product-level inflation 1972-1979 by 2-digit SITC

petrol inflation in 1973 was at least an order of magnitude larger than any other non-OPEC oil crisis year (the 2nd OPEC oil crisis takes place in 1979-1980). However, Apparel (SITC 84) is the 2nd largest US import at around 14% of US imports and this appears to have had much more substantial inflation than petrol did in 1973.

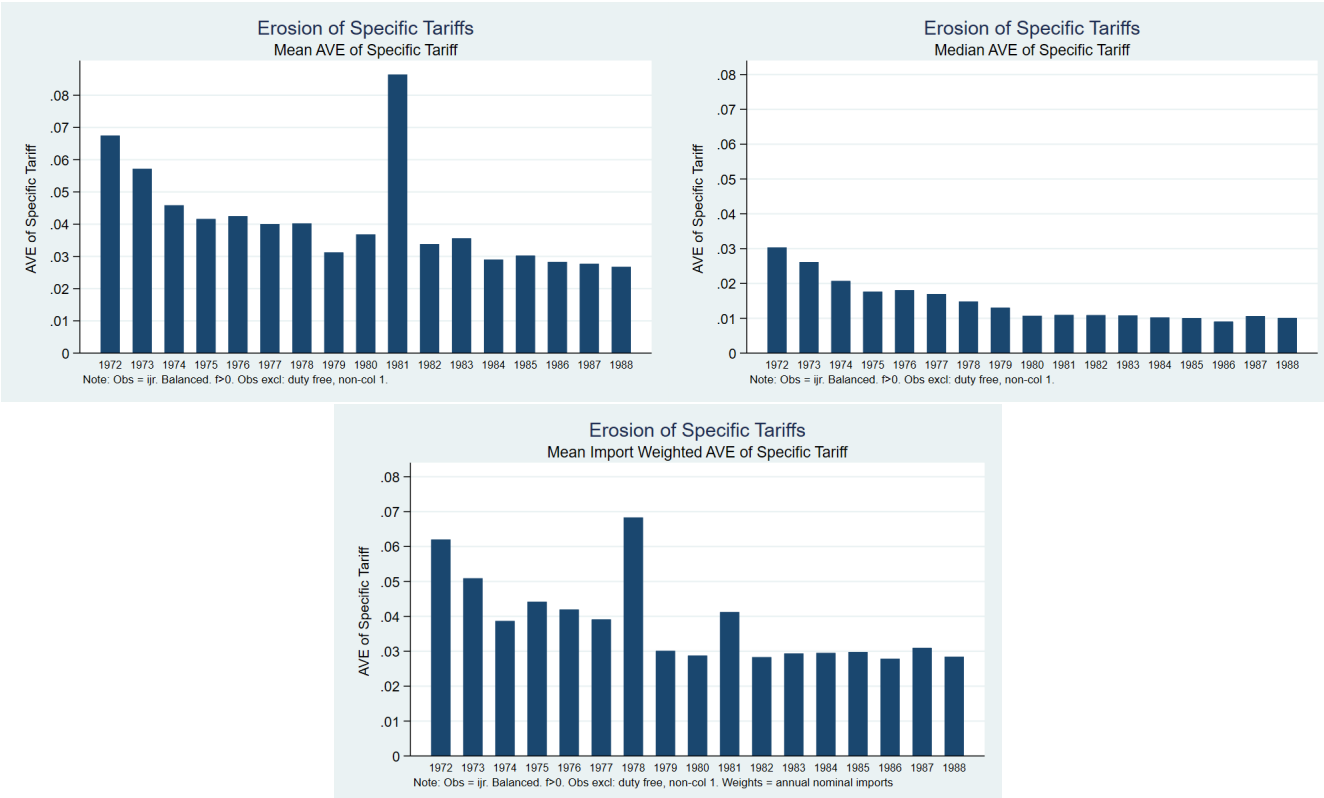

Figure 7: Inflation erosion of specific tariffs

3.3 Changing reliance on specific tariffs

An important factor driving the magnitude of the pass through elasticity of foreign price shocks is the specific tariff share (STS). When a product has no specific tariffs, $STS_{ij} = 0$, then a (positive) 1% foreign price shock passes through completely to the US price as a (positive) 1% price shock. But, pass through will be less than complete in the presence of specific tariffs because inflation erodes the ad valorem equivalent protection afforded by specific tariffs and decreases the overall price paid by the importer.

Prior literature has documented that specific tariff duties accounted for less than 40% of total US tariff duties by the start of the 1970s. Panel (a) Figure 9 shows that this share was around 26% by 1972 once the Kennedy round of GATT liberalization was phased-in. After fluctuating in the 15-25% range leading up to the Tokyo Round, the aggregate specific tariff share falls into the 5-10% range from 1980 onwards.

Importantly, fluctuations in the aggregate STS over our sample period are heavily driven by changes in the product composition – at both the intensive and extensive margin – of

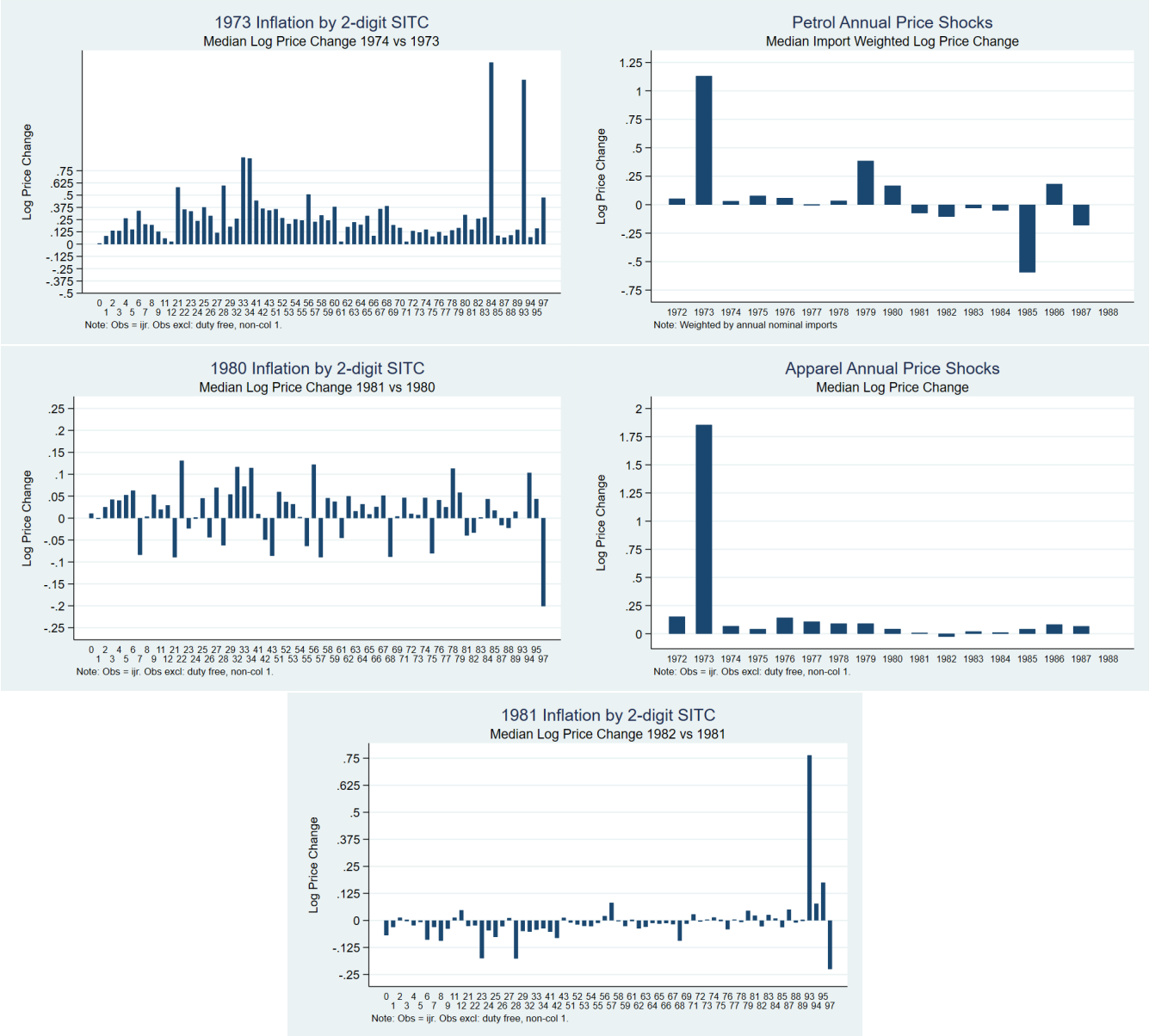

Figure 8: Examples

US imports. Panel (b) of Figure 9 shows the mean STS across the subset of 5-digit TSUSA products imported every year in our sample. Here, we see a stark split between the pre- and post-Tokyo Round periods. The STS is extremely stable at around 32% before the Tokyo Round. But the STS drops sharply to 25% upon implementation of the Tokyo round tariff cuts and continues falling slowly to around 22.5% by 1988. This indicates a dramatic conversion of specific tariffs to ad valorem tariffs at the Tokyo round and a more subtle subsequent phase-in of Tokyo Round tariff cuts.

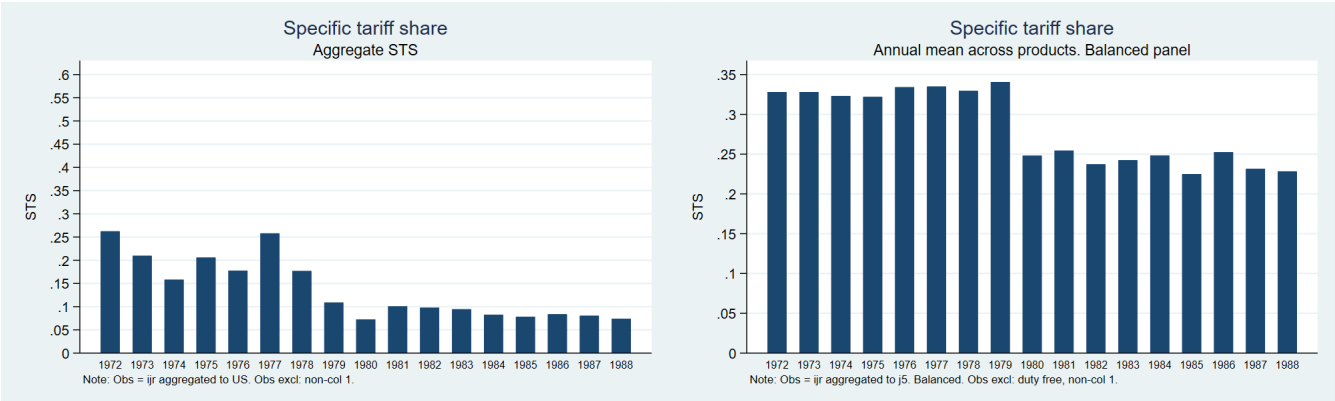

Figure 9: Specific tariff share of total duties

This dramatic change in the reliance on specific tariffs around the Tokyo round is evident more directly in the share of products with specific tariffs. In the full sample of products, panel (a) of Figure 10 shows the share of US imported products with specific tariffs fluctuates between 29% and 34% leading up to the Tokyo Round. Given legislated specific tariffs are essentially constant during this period, these fluctuations represent extensive margin changes in the set of 5-digit products imported by the US. The share of products with specific tariffs and then drops immediately to the 18%-21% range upon implementation of the Tokyo Round. Looking at the balanced panel set of US imported products to remove extensive margin effects, panel (b) shows the share of products with specific tariffs is extremely stable at 32% before the Tokyo Round and then plummets to 21% upon implementation of the Tokyo Round. Ultimately, a key stylized fact emerging from our analysis is the dramatic conversion of specific to ad valorem tariffs in the Tokyo Round.

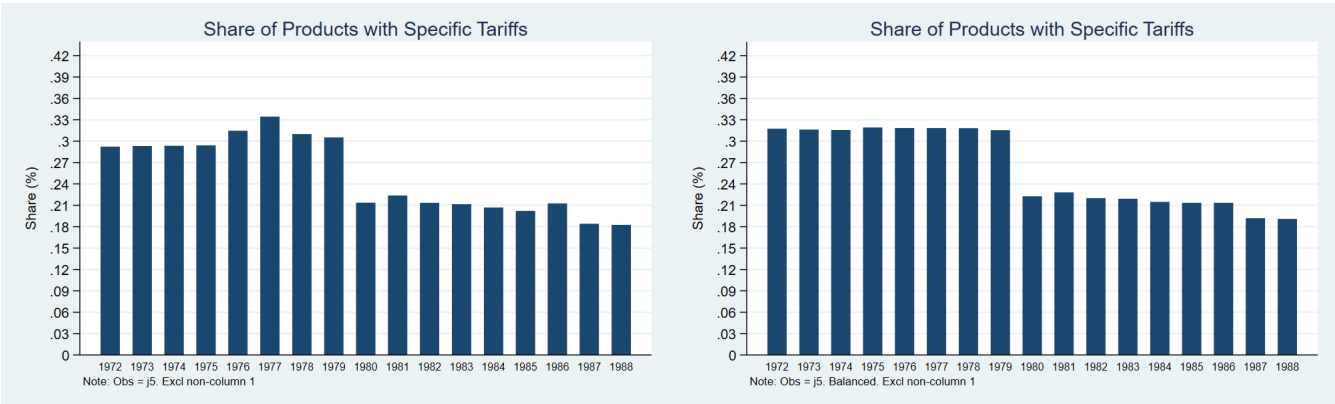

Figure 10: Share of varieties & products with specific tariffs

Figure 11 shows how sectors differed in their reliance on specific versus ad valorem tariffs

and how this reliance changed due to the massive Tokyo Round conversion of specific to ad valorem tariffs. Broadly, 1-digit SITC sectors related to agricultural, minerals and fuels (SITC 0-4 and 9) rely relatively strongly on specific tariffs and, over our sample period, account for about 40% of US imports. In contrast, 1-digit SITC sectors related to chemicals and manufacturing (SITC 5-8) rely relatively strongly on ad valorem tariffs and, over our sample period, account for about 60% of US imports. Thus, there is a clear difference in the kinds of sectors that rely on specific versus ad valorem tariffs.

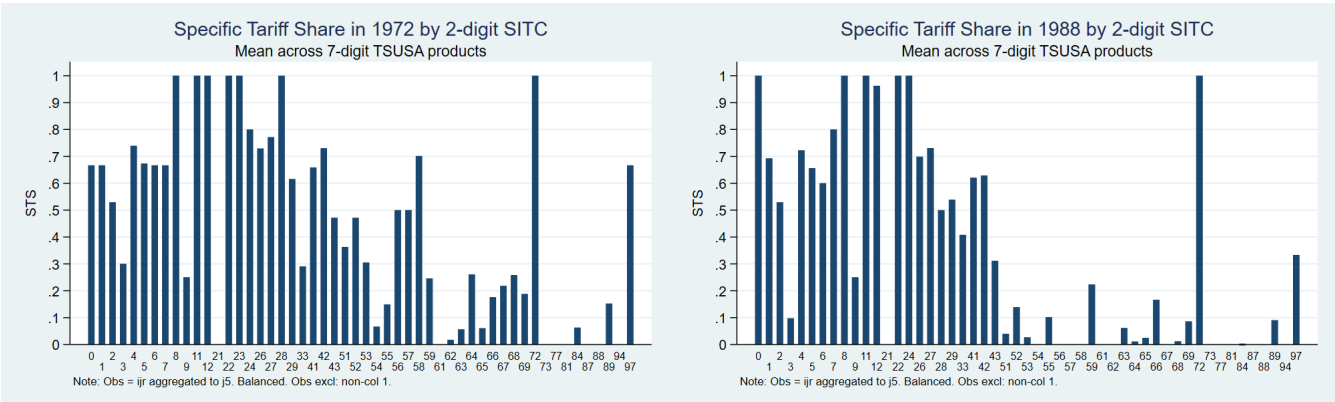

Figure 11: Specific tariff share across sectors

Nevertheless, exceptions do exist. Within the sectors that, overall, rely relatively strongly on ad valorem tariffs, exceptions include pockets of products in chemicals (SITC 5) – such as inorganic chemical (SITC 52), manufactured fertilizer (SITC 56) and artificial resins and plastic materials (SITC 58) – and in manufactured materials (SITC 6) – such as iron and steel (SITC 67) and non-ferrous metals (SITC 68). Indeed, Figure 11 illustrates that the wholesale conversion of specific to ad valorem tariffs in the Tokyo Round focused on removing the specific tariffs that existed in 1-digit sectors that overall relied relatively heavily on ad valorem. Thus, the conversion further strengthened the pre-existing sectoral differences between the sectors that relied relatively heavily on specific tariffs and those that relied relatively heavily on ad valorem tariffs.

3.4 Time-series decomposition of ad valorem equivalence

The ad valorem equivalent tariff for the US as a whole at a given time, AVE_t, is often defined as total duties TD_t as a share of total imports M_t. Figure 12 shows how AVE_t evolved over our sample period spanning both the 1972-1979 pre-Tokyo Round period and the 1979-1987

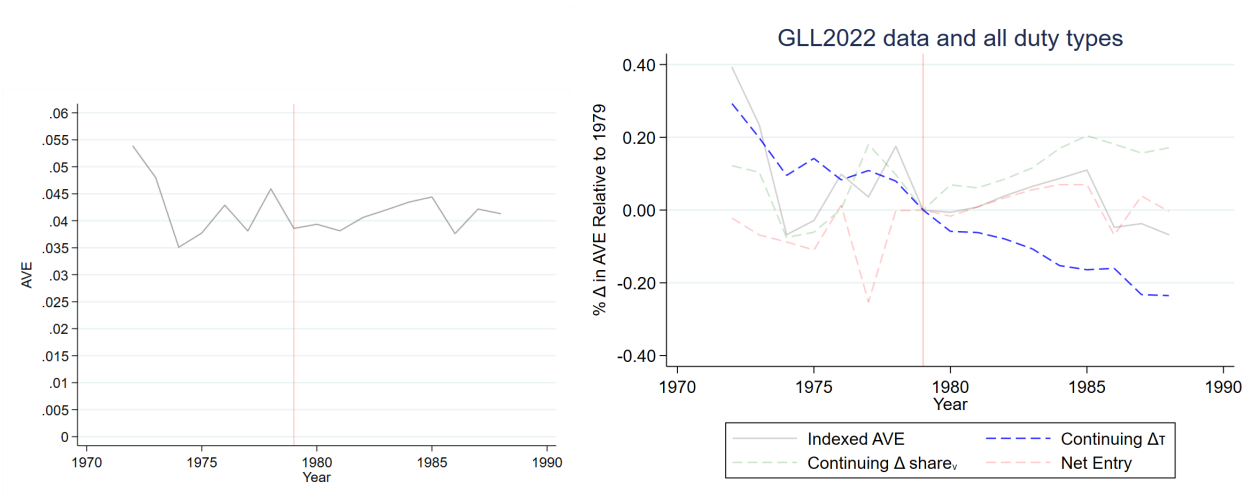

Figure 12: Changing reliance on specific tariffs

post-Tokyo Round period. Two stark facts emerge. First, there was a sharp aggregate liberalization in the early-mid 1970s with the 1972 US AVE sitting around about 30%, or 1.5% points, higher in 1972 than in 1979. Second, there was no significant liberalization thereafter during the phase-in of the Tokyo Round negotiated tariff cuts. Indeed, the US AVE is slightly higher in 1988 than 1979. This is likely uprising since one would naturally expect to see significant AVE liberalization due to the Tokyo Round phasing-in of negotiated tariff cuts.

However, fluctuations over time in the aggregate AVE can happen for various reasons. These reasons include changes in legislated tariff rates, changes in prices which drive changes in the ad valorem equivalence of specific tariffs, changes in the composition of imports across products and varieties, and changes in the set of imported products. Thus, we now explore a decomposition that illustrates these different dimensions of AVE fluctuations.

We can decompose the aggregate importer-level AVE by recognizing

$$\begin{aligned} AVE_{it} &= \frac{\sum_j TD_{ijt}}{\sum_j M_{ijt}} \\ &= \sum_j \frac{M_{ijt}\left(\tau_{ij} + \frac{f_{ij}}{p^*_{ij}}\right)}{\sum_j M_{ijt}} \\ &= \sum_j s_{ijt} AVE_{ijt} \end{aligned}$$

where we defined the variety-level ad valorem tariff equivalence as $AVE_{ij} \equiv \tau_{ij} + \frac{f_{ij}}{p^*_{ij}}$ in (10) and $s_{ijt} = \frac{M_{ijt}}{\sum_j M_{ijt}}$ are time-t import-shares. That is, the importer-level AVE represented by total duties as a share of total imports is equivalent to the import-weighted average of variety level AVE. Given a set of imported varieties J_0 and J_1 in a base-period $t = 0$ and a subsequent period $t = 1$, we can define three mutually exclusive and exhaustive sets of varieties: continuing varieties $J^C = \{j \mid j \in J_0 \text{ and } j \in J_1\}$, a set of exiting products $J^X = \{j \mid j \in J_0 \text{ but } j \notin J_1\}$, and a set of entering products $J^N = \{j \mid j \notin J_0 \text{ and } j \in J_1\}$. Thus, the decomposition of changes in the AVE between two time periods for importer i can be written as

$$\begin{aligned}\Delta AVE_i =& AVE_{i1} - AVE_{i0} \\ =& \sum_{j \in J^C} \left[s_{ij1} \left(AVE_{ij1} - AVE_{ij0} \right) + \sum_{j \in J^C} \left(s_{ij1} - s_{ij0} \right) AVE_{ij0} \right] \\ &+ \sum_{j \in J^N} s_{ij1} AVE_{ij1} - \sum_{j \in J^X} s_{ij0} AVE_{ij0}.\end{aligned}$$

Importantly, the square bracketed term shows that continuing varieties contribute to the AVE both through changes in variety level AVE – due to legislated tariff changes or inflation eroding specific tariffs – and due to changes in the composition of imports across products and varieties. Extensive margin effects also contribute to changes in the AVE through the entering and exiting set of varieties.

Figure 12 shows how each of these dimensions affected the evolution of the US AVE over our sample period at the 5-digit TSUSA level which is the level that the US imposes MFN tariffs. We express changes in the US AVE in percentage terms relative to 1979 because 1979 is the last year before the US phases in its Tokyo Round tariff cut commitments. The grey line shows the AVE bouncing around 3.5% to 4.5% range from 1975 onwards. However, the lack of significant liberalization here is masked by two offsetting effects.

On one hand, we see significant liberalization among the set of continuing varieties when holding the product composition of US imports fixed at 1980 levels. This liberalization (blue line) reflects the inflation erosion of specific tariffs in the pre-Tokyo period and the later phase-in of US Tokyo Round ad valorem tariff commitments. Indeed, it essentially fully accounts for the magnitude – 30% or 1.5% points – of US AVE liberalization in the pre-Tokyo period. Further, as one might expect, we continue to see substantial liberalization in the post-Tokyo Round period with about another 25% drop in the US AVE – around 1%

point – from 1979 to 1987. Thus, the pre- and post-Tokyo Round period was characterized by significant liberalization when holding the composition of US imports fixed.

However, on the other hand, there were substantial changes in the composition of US imports after the Tokyo Round. In particular, among products continually imported, Figure 12 shows that US imports shifted towards relatively higher tariff products (green line). This change in the composition of US imports to higher tariff products essentially completely offsets the substantial liberalization in these products when holding the composition of US imports fixed.

Perhaps surprisingly, the extensive margin dimension plays little role in explaining the evolution of the US AVE. Putting aside the year 1977, the net entry of products can push the US AVE either up or down relative to the 1979 US AVE and never by more than 10%. Moreover, it has zero effect on the US AVE between 1979 and 1988.

Figure 13 shows the distribution of AVE across 2-digit SITC sectors and how this distribution changed between the first and last years of our sample (1972 and 1988). In 1972, the most protected sectors have an AVE in the range of 10-20%. And, they span those that tend to rely on ad valorem tariffs – such as clothing and apparel (SITC 84), handbags and suitcases (SITC 83), footwear (85), photographic equipment and watches (SITC 88) and non-metallic mineral manufacturing n.e.s. (SITC 65) – and those that tend to rely on specific tariffs – such as tobacco (SITC 12), beverages (SITC 11), artificial resins and plastics (SITC 58) and dairy (SITC 2). By 1988, the most protected sectors now have an AVE in the range of 6-12%. Even though all sectors saw significant liberalization, the most protected ones are now sectors that rely very little if at all on specific tariffs – clothing and apparel (SITC 84), handbags and suitcases (SITC 83), footwear (85), and non-metallic mineral manufacturing n.e.s. (SITC 65).

The mean change in the AVE within a 2-digit SITC sector in the pre- and post-Tokyo periods also emphasizes the different types of liberalization in the two periods. Figures 11 and 14 show that pre-Tokyo Round liberalization was concentrated in the agricultural and crude materials sectors which tended to use specific rather than ad valorem tariffs with notable examples including fruit & vegetables (SITC 5), sugar (SITC 6), tobacco (SITC 12), oil seeds (SITC 22) and textile fibers (SITC 26). In contrast, post-Tokyo liberalization was concentrated in the manufacturing sector that tended to use ad valorem rather than specific tariffs with notable examples including textile yarn (SITC 65), apparel and clothing (SITC 84), scientific equipment (SITC 87) and photographic equipment and watches (SITC 88).

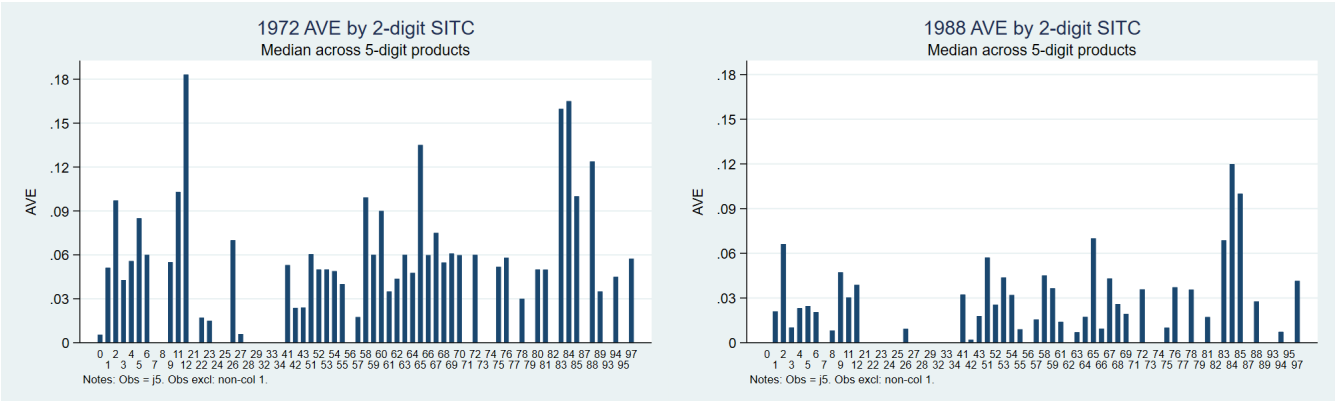

Figure 13: Changing reliance on specific tariffs

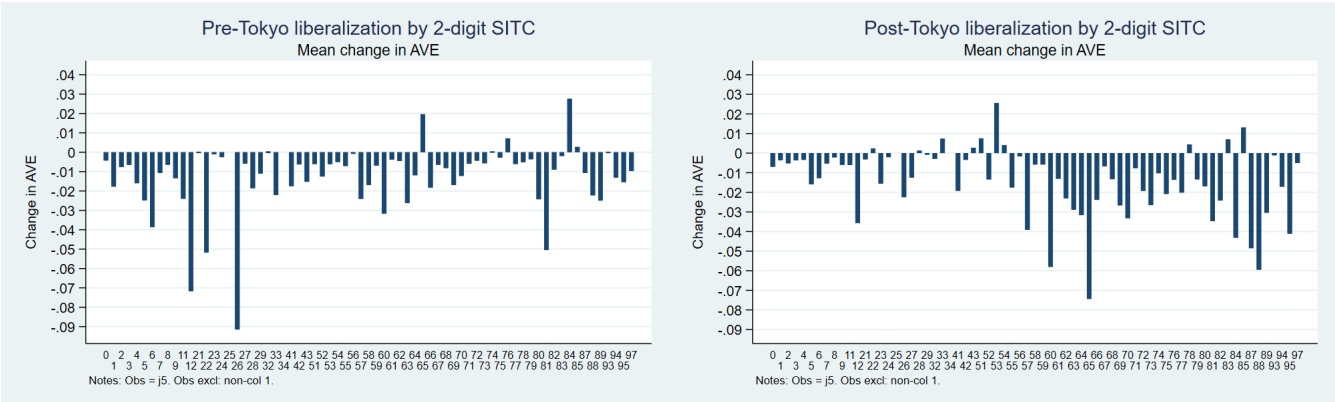

Figure 14: Changing reliance on specific tariffs

An exception that emphasizes the wholesale conversion of specific to ad valorem tariffs is explosives which had a very large liberalization post-Tokyo and saw the specific tariff share of its overall tariff fall from 50% to zero.

3.5 Changes in foreign price shock pass-through elasticity

Section 3.1 showed that the pass through elasticity of a foreign price shock into the imported price of a product depends on both the share of the product's tariffs that are specific (i.e. STS) and the product's ad valorem equivalent tariff. Specifically, equation (11) showed that the term $STS\frac{AVE}{1+AVE}$ exactly captures the extent that the pass-through elasticity is less than complete (i.e. less than 1). We refer to this term as pass through elasticity leakage.

Figure 15 shows how the pass through leakage varies across 2-digit SITC sectors in the first and last years of our sample (1972 and 1988). Among 5-digit products with specific

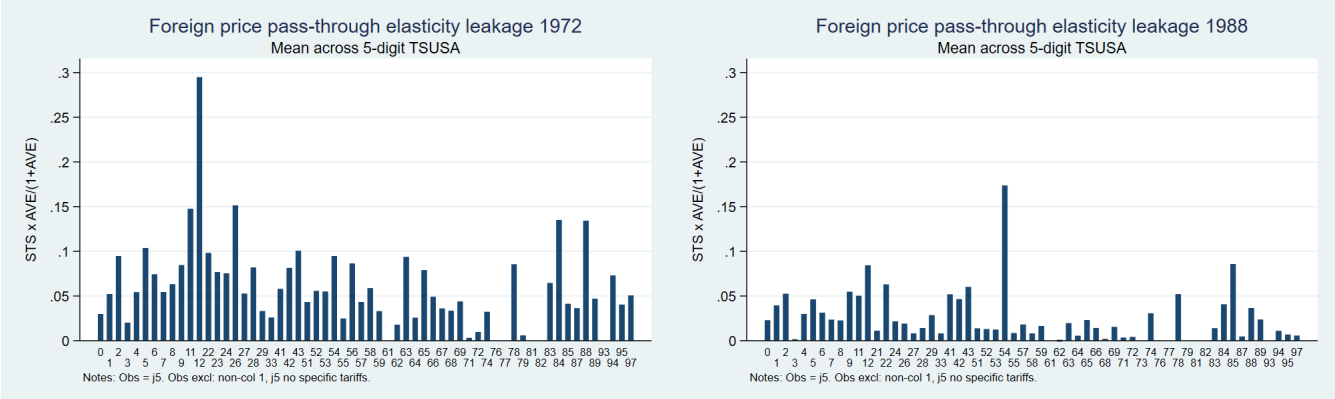

Figure 15: Changing foreign price shock pass through

tariffs in 1972, the mean pass through leakage was .06. Thus, in percentage terms, the pass through elasticity was 94%. Certain types of raw materials (e.g. tobacco stems, tobacco leaves, amber and wool), have pass through leakage in the range of .55-.80 range. These are products in sectors that have relatively high STS and hence tend to rely heavily on specific tariffs (beverages (SITC 11), tobacco (SITC 12) and textile fibers (SITC 26)). However, some sectors that rely quite little on specific tariffs (i.e. low STS) also have a high pass through leakage because of their very high protection (i.e. high AVE). Clothing and apparel (SITC 84) and photographic equipment and watches (SITC 88) both have mean pass-through leakage around 10% with leakage at the 5-digit TSUSA level on some products – e.g. leather gloves, lace underwear, watches, and clocks – in the 20-50% range. Thus, pass through leakage can notably affect products with relatively high or relatively low reliance on specific tariffs.

Naturally, the three distinct types of liberalization during our sample – pre-Tokyo Round inflation erosion of specific tariffs, the Tokyo Round conversion of specific to ad valorem tariffs, and the subsequent phase out of ad valorem tariffs – dramatically shrink the pass through leakage by 1988. Across 5-digit TSUSA products with specific tariffs, the mean leakage is now only 67% lower at .02 and the number of products with leakage above 20% has fallen by more than 80%. Given this broad liberalization and lowering of the AVE, the products among the highest pass through leakage tend to be those that largely retained their non-trivial level of specific tariffs. These sectors include beverages (SITC 11), tobacco (SITC 12), oil seeds (SITC 22), animal oils and fats (SITC 41), vegetable oils and fats (SITC 42), and processed animal and vegetable oils and fats (SITC 43). But, the pass through leakage is still notably lower for these sectors in 1988 than in 1972.

4 Conclusion

Inflationary foreign price shocks reduce the ad valorem equivalence of US specific tariffs. We document that such inflationary prices shocks generated a greater liberalization of ad valorem equivalent US applied tariffs over the 1972-1979 period than the Tokyo Round over the over the 1980-1988 period. To do so, we create the first database of legislated ad valorem and specific tariffs in the US tariff code for the 1972-1988 period. We show that our procedure for estimating US tariffs in this period is more accurate than a *perfect* digitization of the US tariff code from PDF files because of a frequent discrepancy in the unit of quantity measurements between the US tariff code and the official US import data that only becomes apparent due to our estimation procedure.

Our analysis thus far leaves plenty of avenues for current and future research. The dramatic "accidental" liberalization US ad valorem equivalent tariffs through inflationary foreign price shocks reducing should have notable impacts on US import growth, US output, and US labor markets. Our theoretical result that the pass through elasticity of foreign price shocks to US import prices depends on specific tariffs also has implications for the estimation of trade elasticities. This pass through result also implies that the 1972-1988 period is one where both the mean and volatility of US import prices is changing due to US tariff shocks. In turn, it provides a natural setting to explore issues related to the first-moment and second-moment effects of trade shocks on welfare and labor market outcomes. Finally, the per unit nature of specific tariffs and existing literature linking quality to per unit trade costs leads to interesting issues involving endogenous quality.

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